Math Diagnostic — Practice Test

Algebra & Functions · timed diagnostic format

32	46	~86 sec	No
questions	minutes	per question	calculator

About this test

- This is the timed math diagnostic from your photos, retyped so it's clean and printable. Wording, numbers and answer choices are kept as they appeared on screen.
- The original had 33 questions with a 46-minute timer. One question (number 7 or 8 in the original) wasn't in the photos, so this version has 32, numbered 1–32 in the original order.
- Graphs are redrawn on a clean grid to match what was on screen.
- Multiple choice, A–D. No calculator. No penalty for guessing — answer everything.

How to take it

- Set a timer for 46 minutes and don't stop it.
- Skip anything that takes more than about 90 seconds and come back to it.
- The answer key with a short explanation for every question is at the back.

What's on it

Simplifying, expanding & factoring	Questions 1, 5, 17, 27
Number series	Questions 3, 6, 14, 23, 28
Plugging in & solving formulas	Questions 8, 11, 12, 13, 22, 24, 25
Isolating a variable	Questions 19, 20, 21
Variable relationships (which statement is true)	Questions 2, 9, 10, 26, 30, 32
Same relationship: tables, graphs, words	Questions 4, 7, 15, 29
Graphing & systems	Questions 16, 18, 31

Question 1

Simplify the following expression to its simplest form.

Consider $P = 3$ and $Q = -2$:

$$Q(3P^2 - 2P) + (Q - P)^2$$

A) -17

B) $P^3 - Q^2 + 13$

C) $-15 + P^2$

D) -32

WORKSPACE

Question 2

Consider the following formula:

$3T = Q + 3$

Which of the following statements is true for this formula?

A) For negative Q values, $T < 0$.

B) For positive T values, $Q > 0$.

C) The difference between the values of T and Q is constant.

D) The difference between the values of T and Q changes.

WORKSPACE

Question 3

What is the next number in the following series of numbers?

68 46 29 17 ?

A) 7

B) 10

C) 12

D) 13

WORKSPACE

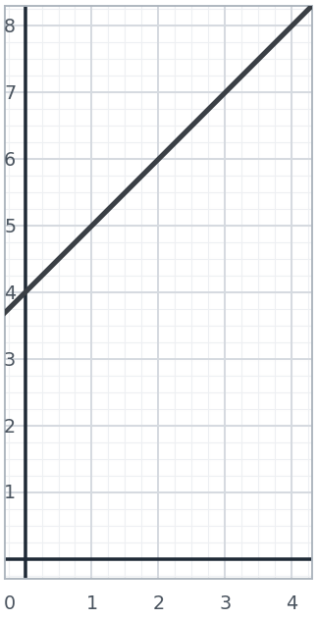
Question 4

Consider the following table, which describes some coordinates of a linear function:

x	y
4	0
8	1
12	2
16	3
20	4

Which of the following choices represents the same relationship as the given table?

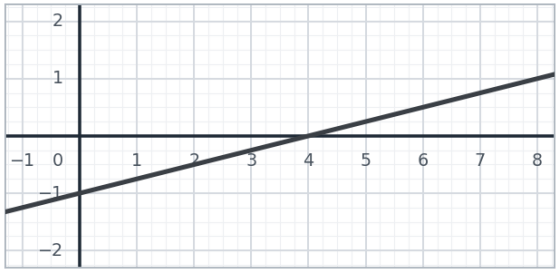
A)



B)

x	y
0	−4
1	−3
2	−2
3	−1
4	0

C)



D) Y is equivalent to four times the value of X and a constant C, where C equals 4.

WORKSPACE

Question 5

Consider the following formula:

$y = 3(x + 2)(2x - 3)$

Which of the following formulas is equivalent to this one?

A) $6x^2 + 3x - 18$

B) $6x^2 + 3x - 9$

C) $3x^2 + 6x - 18$

D) $3x^2 + 3x - 6$

WORKSPACE

Question 6

What is the next number in the following series of numbers?

9 15 12 17 14 18 15 ?

A) 13

B) 18

C) 16

D) 19

WORKSPACE

Question 7

Consider the following function:

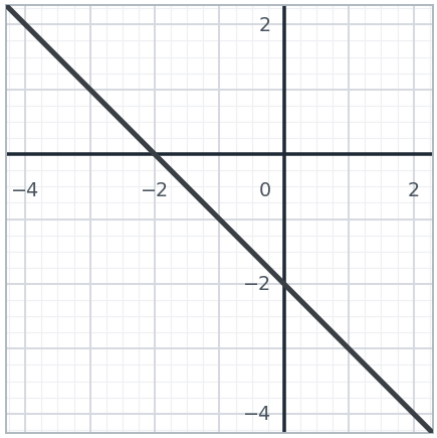
$y = 2x - 3$

Which of the following choices represents the same relationship as the given function?

A)

x	y
−1	−5
0	−3
1	−2
2	−1
3	0

B)



C)

x	y
4	5
5	7
6	9
7	11

D) Y is equal to twice the value of X subtracted by three times the value of X.

WORKSPACE

Question 8

Consider the following formula:

$$(4A - (B + 3C)) + 10 = 2BA - C$$

If $A = 3$ and $C = 4$, what is the value of B?

A) 0

B) 2

C) 4

D) 3

WORKSPACE

Question 9

Consider the following formula:

$x = 4y + 8$

Which of the following statements is true for this formula?

A) When the value of y is less than -2, x is negative.

B) When the value of x is less than 10, y is negative.

C) When the value of x is less than 2, y is positive.

D) When the value of y is greater than -4, x is positive.

WORKSPACE

Question 10

Consider the formula given:

$C = 10 - D$

Which of the following conclusions cannot be drawn from this formula?

A) If C is greater than 10, D is negative.

B) If D is greater than -10, C is positive.

C) If C is less than -10, D is positive.

D) If D is greater than 10, C is negative.

WORKSPACE

Question 11

Given the equation:

$M(N-4) = 67 - (N + D(N+3))$

If $D = 2$ and $M = 4$, what is the value of N?

A) 5

B) 9

C) 11

D) 13

WORKSPACE

Question 12

What is the value of this expression?

For $z = 8$ and $w = \frac{2z}{4}$,

$$\frac{3z+2}{5w-7} + \frac{5w-4}{3z} = ?$$

A) $2\frac{2}{3}$

B) $2\frac{5}{8}$

C) $3\frac{1}{5}$

D) $3\frac{1}{3}$

WORKSPACE

Question 13

Evaluate the value of the unknown variable:

$$\frac{6e-4f}{3g+h}=ab$$

$a=6,\ e=-4,\ f=9,\ g=4$, and $h=-7$. Find b.

A) -2

B) -12

C) -6

D) -18

WORKSPACE

Question 14

What is the next number in the following series of numbers?

1.3 3.9 11.7 35.1 ?

A) 78.3

B) 98.2

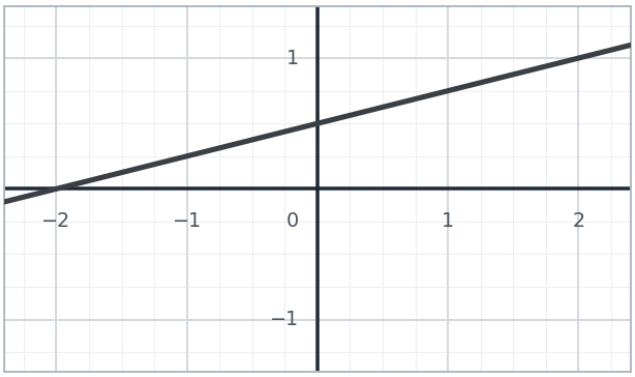
C) 105.3

D) 115.6

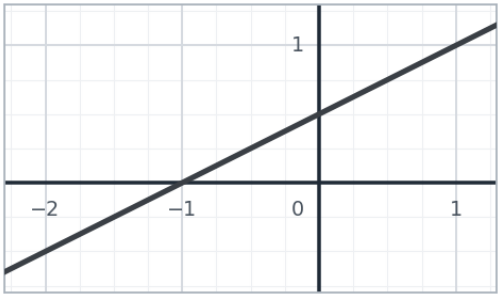
WORKSPACE

Question 15

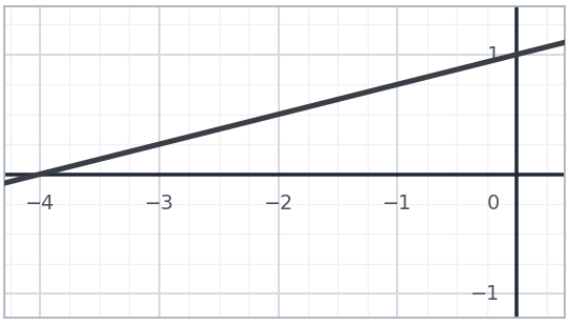
Consider the following graph of the function $y = mx + b$:



Graph A



Graph D



Which of the following options is true in regard to this graph?

A) Graph A is of a function with the same slope m and the same interception point with the y-axis.

B) Function B, $y = 0.25x + 0.25$, has a different slope and a different interception point with the y-axis.

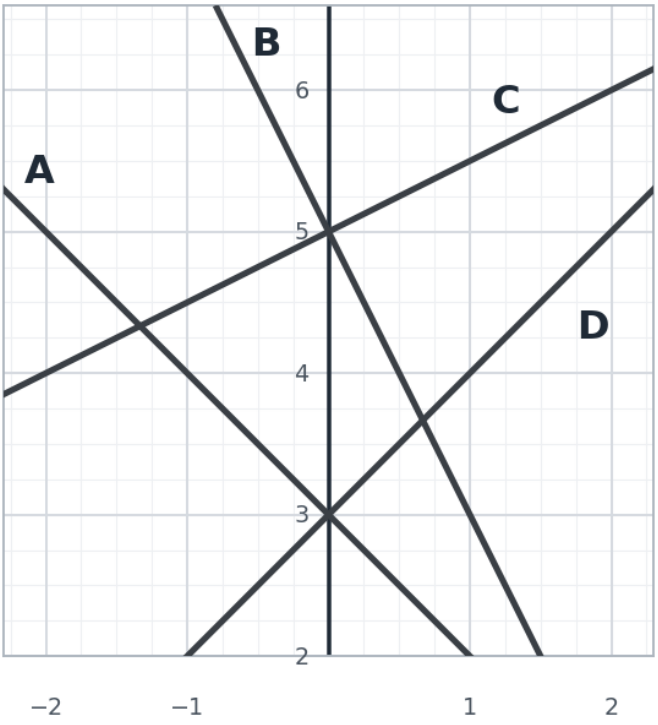
C) Function C, $y = 0.5x + 0.5$, has the same slope and the same interception point with the y-axis.

D) Graph D is of a function with the same slope m and a different interception point with the y-axis.

WORKSPACE

Question 16

Which of the following graphs represents the function $y = -2x + 5$?



A) A

B) B

C) C

D) D

WORKSPACE

Question 17

Factor the polynomial:

$x^2 + 3x - 28$

A) $(x - 4)(x + 7)$

B) $(x + 6)(x - 3)$

C) $(x + 3)(x - 6)$

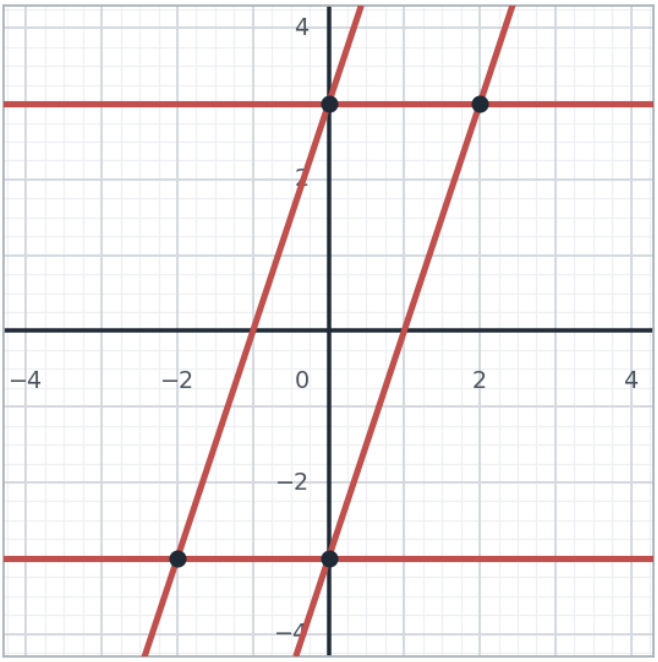
D) $(x - 7)(x + 4)$

WORKSPACE

Question 18

Which point on the graph represents the intersection point of the functions:

1. $y = 3x - 3$ 2. $y = 3$



A) $(-2, -3)$

B) $(0, 3)$

C) $(0, -3)$

D) $(2, 3)$

WORKSPACE

Question 19

Isolate the variable from the formula:

$$A = \frac{1}{2}h(b_1 + b_2)$$

$b_1 = ?$

A) $\frac{2Ah}{b_2}$

B) $\frac{2A}{h} - b_2$

C) $\frac{2Ab_2}{h}$

D) $\frac{1}{2}Ah - b_2$

WORKSPACE

Question 20

Isolate the variable from the formula:

$S = r\theta$

$\theta = \frac{d}{\pi r}$

$S = ?$

A) $\frac{\pi r^2}{d}$

B) $\frac{d}{\pi}$

C) $\frac{\pi d}{r}$

D) $\frac{\pi d}{r^2}$

WORKSPACE

Question 21

Isolate the variable from the formula:

$$\frac{7x-12}{4}=y$$

$x=?$

A) $\frac{4}{7}y+12$

B) $7y-12$

C) $\frac{4}{7}y+\frac{12}{7}$

D) $7(4y+12)$

WORKSPACE

Question 22

Solve the linear equation:

$18x + 7 = -12x + 22$

A) 0.5

B) 2.5

C) -0.5

D) -2.5

WORKSPACE

Question 23

Identify the next term in the series presented below:

2 5 10 17 ?

A) 20

B) 24

C) 26

D) 28

WORKSPACE

Question 24

What is the solution of this proportion?

$$\frac{m-2}{15+m}=\frac{1}{52}$$

A) $\frac{5}{3}$

B) $\frac{7}{3}$

C) $\frac{4}{3}$

D) $\frac{8}{3}$

WORKSPACE

Question 25

Solve the following system of equations:

i. $4x + 2y = 18$

ii. $6x - y = 13$

A) (2.75, 3.5)

B) (2.5, 4)

C) (2.5, 2)

D) (3.5, 2)

WORKSPACE

Question 26

Consider the following formula:

$Q = W - 1$

Which of the following statements is true for this formula?

A) $W + Q > 0$

B) $Q + W > 2Q$

C) $W - Q < 0$

D) $(Q - W) > (W - Q)$

WORKSPACE

Question 27

Simplify the following expression:

$3(2x^3 - 4x^2 + 5 - x + 3x^2 - 5x^3) + 3x^3 + 3x^2 + 3x + 3$

A) $-6x^3 - 3x^2 - 3x + 15$

B) $-6x^3 + 18$

C) $6x^2 + 6x + 24$

D) $-6x^2 - 3x + 15$

WORKSPACE

Question 28

What is the next number in the following series of numbers?

9 10 13 22 49 ?

A) 76

B) 81

C) 103

D) 130

WORKSPACE

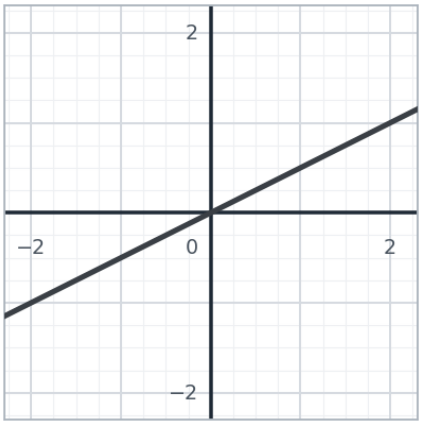
Question 29

Consider the following information:

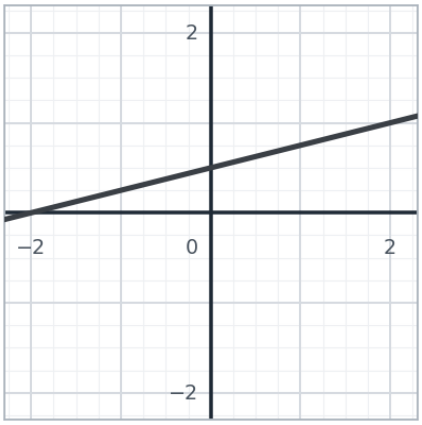
Y is equivalent to four times the value of X divided by eight.

Which of the following choices represents the same relationship as the aforementioned information?

A)



B)



C)

x	y
−4	2
−8	4
−12	6
−16	8
−20	10

D)

x	y
0	0
2	1
4	3
6	4
8	6

WORKSPACE

Question 30

Consider the following formula:

$A = 0.25B + 4.5$

Which of the following statements is true for this formula?

A) When the value of B is greater than -18, A is negative.

B) When the value of B is lower than -18, A is positive.

C) When the value of B is greater than -9, A is positive.

D) When the value of B is lower than -9, A is negative.

WORKSPACE

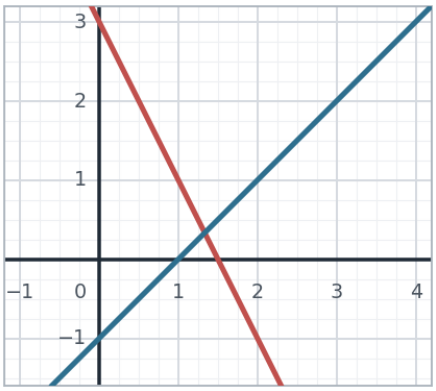
Question 31

Which of the graphs represents the following two linear equations?

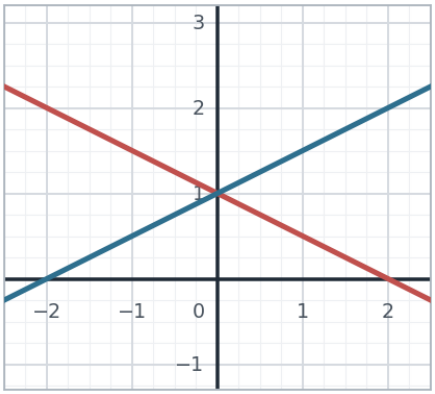
$2x - y = 4$

$x + y = 2$

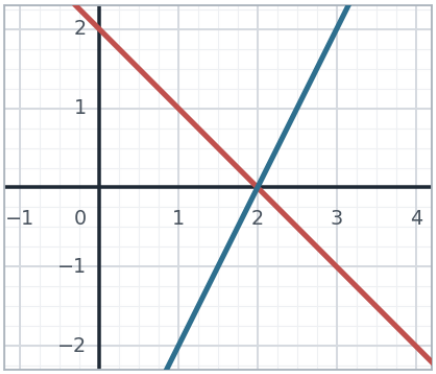
A)



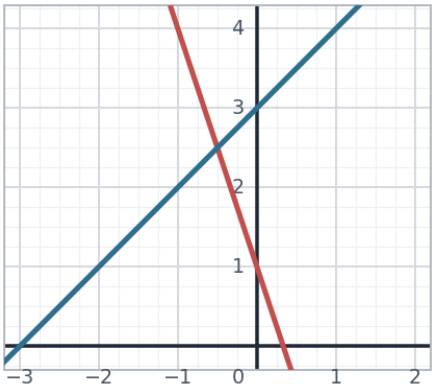
B)



C)



D)



WORKSPACE

Question 32

Consider the following formula:

$$\frac{1}{2}y = 2 - \frac{1}{3}x$$

Which of the following statements is true for this formula?

A) When the value of y is less than 4, x is negative.

B) When the value of y is greater than 4, x is positive.

C) When the value of x is less than -6, y is positive.

D) When the value of x is greater than -6, y is positive.

WORKSPACE

Answer Key

1. A	**9. A**	**17. A**	**25. A**
2. D	**10. B**	**18. D**	**26. B**
3. B	**11. C**	**19. B**	**27. B**
4. C	**12. A**	**20. B**	**28. D**
5. A	**13. A**	**21. C**	**29. A**
6. B	**14. C**	**22. A**	**30. C**
7. C	**15. D**	**23. C**	**31. C**
8. B	**16. B**	**24. B**	**32. C**

1. A $-2(27-6)+(-2-3)^2=-42+25=-17$

2. D $T=\frac{Q+3}{3}$, so $T-Q=1-\frac{2}{3}Q$, which changes as Q changes.

A fails at $Q=-1$ ($T=\frac{2}{3}$). B fails at $Q=-2$ ($T=\frac{1}{3}$).

3. B Differences: -22, -17, -12 (each 5 smaller). Next: -7. $17-7=10$

4. C y goes up 1 every time x goes up 4, and $y=0$ at $x=4$: $y=\frac{1}{4}x-1$

Graph C crosses at (0, −1) and (4, 0).

5. A $(x+2)(2x-3)=2x^2+x-6$. Times 3: $6x^2+3x-18$

6. B $+6, -3, +5, -3, +4, -3$ … next is $+3$: $15+3=18$

7. C $2(4)-3=5$, $2(5)-3=7$, $2(6)-3=9$, $2(7)-3=11$ ✓

Table A works for the first two rows, then breaks at $x=1$.

8. B $12-B-12+10=6B-4 \rightarrow 10-B=6B-4 \rightarrow 7B=14 \rightarrow B=2$

9. A $x=0$ when $y=-2$. Below -2, x is negative.

D fails at $y=-3$ ($x=-4$).

10. B C is positive only when $D<10$. $D=15$ is greater than -10 but gives $C=-5$.

So B is the one that can't be concluded.

11. C $4N-16=67-N-2N-6 \rightarrow 4N-16=61-3N \rightarrow 7N=77 \rightarrow N=11$

12. A $w=\frac{16}{4}=4$. $\frac{26}{13}+\frac{16}{24}=2+\frac{2}{3}=2\frac{2}{3}$

Answer Key (cont.)

13. A $\frac{-24-36}{12-7}=\frac{-60}{5}=-12.\quad -12=6b \;\rightarrow\; b=-2$

Trap: -12 is the left side, not b.

14. C Each term is 3 times the last: $35.1\times 3=105.3$

15. D Given line: crosses y at 0.5 and x at $-2 \;\rightarrow\; y=0.25x+0.5$

Graph D: $y=0.25x+1$ — same slope, different y-intercept ✓

Graph A and C have slope 0.5.

B has the same slope (0.25), so 'different slope' is false.

16. B Look for the line crossing the y-axis at 5 and falling 2 for every 1 right.

B and C both cross at 5; only B falls (C rises).

17. A Need two numbers that multiply to -28 and add to $+3$: $+7$ and -4.

18. D Set them equal: $3x-3=3 \;\rightarrow\; x=2.$ Point: (2, 3).

The graph also shows $y=3x+3$ and $y=-3$ as distractors.

19. B Times 2: $2A=h(b_1+b_2).$ Divide by h: $\frac{2A}{h}=b_1+b_2.$ Subtract b_2.

20. B $S=r\cdot\frac{d}{\pi r}=\frac{d}{\pi}$ (the r's cancel)

21. C $7x-12=4y \;\rightarrow\; 7x=4y+12 \;\rightarrow\; x=\frac{4}{7}y+\frac{12}{7}$

22. A $30x=15 \;\rightarrow\; x=\frac{1}{2}=0.5$

23. C Differences 3, 5, 7 … next is 9: $17+9=26$

24. B $52(m-2)=15+m \;\rightarrow\; 52m-104=15+m \;\rightarrow\; 51m=119 \;\rightarrow\; m=\frac{7}{3}$

25. A From ii: $y=6x-13.$ Plug in: $4x+12x-26=18 \;\rightarrow\; 16x=44 \;\rightarrow\; x=2.75$

$y=6(2.75)-13=3.5$

26. B W is always 1 more than Q, so $Q+W=2Q+1>2Q$ ✓

$W-Q=1$ (not <0); $Q-W=-1$, which is less than 1.

Answer Key (cont.)

27. B Inside first: $-3x^3 - x^2 - x + 5$. Times 3: $-9x^3 - 3x^2 - 3x + 15$.
Add the rest: $-6x^3 + 0x^2 + 0x + 18 = -6x^3 + 18$

28. D Differences 1, 3, 9, 27 (times 3 each time). Next is 81: $49 + 81 = 130$

29. A $y = \frac{4x}{8} = \frac{1}{2}x$: through (0, 0), up 1 for every 2 right. That's graph A.
Table C has the wrong sign; table D breaks at $x = 4$ (should be 2).

30. C $A = 0$ at $B = -18$. Above -18, A is positive — so above -9 it's positive too.
D fails at $B = -10$ ($A = 2$).

31. C Get y alone: $y = 2x - 4$ and $y = -x + 2$.
Both cross the x-axis at (2, 0).
Graph C: blue rises 2 per 1, red falls 1 per 1, and they meet at (2, 0).

32. C Times 2: $y = 4 - \frac{2}{3}x$. $y = 0$ at $x = 6$; y is positive whenever $x < 6$.
So $x < -6 \rightarrow$ positive ✓. D fails at $x = 9$ ($y = -2$).